Wireshark/tshark evidence (real capture)

Captured on lo0 with tshark 4.6.9 during a 60 MB transfer through the emulator (3 paths, path 3 dies at 6 s). Decoded with wireshark/mpft.lua. Commands are shown above each output. These are text extracts, not GUI screenshots.

TCP handshakes (tcp.flags.syn==1)

```
tshark -r mpft.pcapng -Y 'tcp.flags.syn==1' -T fields -e frame.time_relative -e tcp.srcport -e tcp.dstport -e t
0.000000000 49922   9201    False
0.000003000 49923   9202    False
0.000005000 49924   9203    False
0.000046000 9201    49922   True
0.000061000 9202    49923   True
0.000076000 9203    49924   True
6.210553000 49955   9203    False
6.413139000 49956   9203    False
6.818631000 49957   9203    False
7.620042000 49958   9203    False
```

MPFT messages per type (mpft.type)

```
tshark -r mpft.pcapng -Y mpft -T fields -e mpft.type | sort | uniq -c   (1 HELLO, 2 MANIFEST, 3 MANIFEST_ACK, 4
   5 1
   1 1,2
   1 2
   2 3
 459 4
 458 5
   4 7
```

Decoded MANIFEST (first, truncated)

```
MPFT MANIFEST
    Magic: MP
    Version: 1
    Type: MANIFEST (2)
    Session ID: 0xe3a2e66d
    Chunk index: 0
    Payload length: 15755
    Header CRC32: 0xa90d94bb
    JSON payload […]: {"name": "video.bin", "size": 60000000, "chunk_size": 262144, "total_chunks": 229, "file_
```

Decoded CHUNK header fields

```
frame.number    tcp.dstport mpft.session    mpft.chunk  mpft.len    mpft.crc
96  9201    0xe3a2e66d  0   262144  0x1e63f1ac
114 9203    0xe3a2e66d  1   262144  0x2303d81c
126 9202    0xe3a2e66d  2   262144  0x64a3a2cc
251 9000    0xe3a2e66d  0   262144  0x1e63f1ac
```

Bytes per path (TCP conversations, sender-to-proxy ports)

```
Filter:<No Filter>
127.0.0.1:49922           <-> 127.0.0.1:9201               725 43 kB        2472 35 MB        3197 35 MB
127.0.0.1:49923           <-> 127.0.0.1:9202               271 16 kB        1085 15 MB        1356 15 MB
127.0.0.1:49924           <-> 127.0.0.1:9203               166 9988 bytes     635 9211 kB       801 9221 kB
127.0.0.1:49955           <-> 127.0.0.1:9203                 1 44 bytes        1 68 bytes        2 112 byte
127.0.0.1:49956           <-> 127.0.0.1:9203                 1 44 bytes        1 68 bytes        2 112 byte
127.0.0.1:49957           <-> 127.0.0.1:9203                 1 44 bytes        1 68 bytes        2 112 byte
127.0.0.1:49958           <-> 127.0.0.1:9203                 1 44 bytes        1 68 bytes        2 112 byte
```

TCP retransmissions (tcp.analysis.retransmission)

```
count: 0
```

Path 3 (port 9203) closing at the kill (FIN/RST)

```
6.004847000 9203    49924   False   True
6.005392000 49924   9203    False   True
6.210572000 9203    49955   True    False
6.413174000 9203    49956   True    False
6.818662000 9203    49957   True    False
7.620070000 9203    49958   True    False
```

Notes: SYNs on port 9203 after 6.0 s are the sender's reconnect attempts after the path was killed; they are refused (RST,ACK) with growing gaps (0.2, 0.4, 0.8 s), then the path is declared down. The count of 0 TCP retransmissions is expected on loopback. A line such as `1,2` in the type counts means two MPFT messages (HELLO and MANIFEST) were in one TCP segment.